

Lab: Brain Processing -- How Your Neurons Think

Objective

Learning Goal: Investigate how your brain processes information by exploring neural pathways, brain regions, and how the nervous system updates its internal model.

This is an **investigation lab** designed for middle school students (Grades 6-8).

Materials Needed

- Printed brain diagram or poster (side view showing major regions)
 - Colored pencils or markers
 - Index cards for neuron chain activity
 - Timer or stopwatch
 - Lab journal or notebook
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Part 1: Brain Region Mapping (10 min)

Different parts of your brain handle different types of thinking. Map the major regions and their jobs.

Brain Region	Location	What it handles	Example from your life
Prefrontal cortex	Front	Planning, decisions, self-control	
Amygdala	Deep center	Emotions, fear responses	
Hippocampus	Inner temporal lobe	Memory formation	
Cerebellum	Back/bottom	Coordination, balance	
Visual cortex	Back	Processing what you see	
Motor cortex	Top center	Controlling movement	

Write your response here

Part 2: Neural Telephone Game (15 min)

Neurons communicate by passing signals along a chain. Let's simulate this.

Activity: Line up 5-6 people. The first person sees a simple image. They whisper a description to the next person, who whispers to the next, and so on. The last person draws what they heard.

Stage	What was communicated	Accuracy (1-10)
Original image	(teacher shows)	10
Person 1 to 2		
Person 2 to 3		
Person 3 to 4		
Person 4 to 5		
Final drawing		

Write your response here

Part 3: Brain Speed Test (10 min)

Your brain processes different types of information at different speeds. Test this with the Stroop Effect.

Stroop Test: Write color words in mismatched ink colors (e.g., the word "RED" written in blue ink). Time how long it takes to name the INK COLOR (not the word) for 10 items.

Condition	Time to complete 10 items	Errors
Matching (word and color agree)		
Mismatching (word and color disagree)		

Write your response here

Part 4: Reflection Table

Question	Your Answer
What are the major brain regions and what do they do?	
How do neurons pass information along a chain?	
What does the Stroop Effect reveal about brain processing?	
How does the brain update its model when processing conflicts?	
How does Active Inference explain brain cognition?	

Write your response here